

SIMONA GURRI'

Email ◇ LinkedIn ◇ Homepage ◇ Google Scholar

Researcher with a focus on applied Computational Fluid Dynamics and horizontal knowledge in transport systems: leveraging advanced simulation techniques to drive innovation in sustainable mobility and energy efficiency. Loving to disseminate research results. Speaking Italian, English, French and a bit of Spanish, learning Russian and German.

POSITION

Politecnico di Torino, Italy

RESEARCH FELLOW

Energy engineering - Computational Fluid Dynamics - Fluid Machinery

Advisor: Alessandro Ferrari

AFFILIATION

2023 - present

Dept. of Energy (DENERG) - Politecnico di Torino

2021 - 2023

Dept. of Environment, Transport and Infrastructure (DIATI) - Politecnico di Torino

INTERESTS

Topics

Simulation and modeling · Powertrain · Transportation systems

Methods

Reactive Computational Fluid Dynamics · Time Frequency Analysis · Model Based System Engineering · Discrete Event Simulation · Microsimulation

EDUCATION

Ph.D. in Energetics

Italy

Politecnico di Torino

2021 - 2025

- Awarded **cum laude**. Thesis: "Synergies between Bimodal Freight Electric Multiple Units Trains and Advanced ICE Technologies for Sustainable Intermodal Transport" Supervisors: Ferrari A., Dalla Chiara B.

M.Sc. in Mechanical Engineering (Erasmus+)

France

EPF Sceaux

2019 - 2020

- Research project: "FEM modeling and simulation in python of the sealing processes of a Nitinol ring for a particle accelerator" Supervisor: Belhadj C.
- Courses held in French.

B.Sc. and M.Sc. in Mechanical Engineering

Italy

Politecnico di Torino

2015 - 2021

- Grade: 110/110 · Thesis: "Functional Testing of a passenger train: the case of the Donizetti regional rolling stock– Systems Engineering approach and basis for Remote Prognostics" Supervisors: Dalla Chiara B., Bonamico M.

PROFESSIONAL EXPERIENCE

RESEARCH FELLOW

B4IA project @ Politecnico di Torino

10/2025 - Present

- Fellowship funded by the *Italian Ministry of Economics*.
- Design and prototyping (TRL 6) of an autonomous bus with a **PFI hydrogen-fueled internal combustion engine**. Focus on engine and injection system (**1D CFD** and **3D CFD** simulations for injectors' and combustion chamber design).
- Collaborating with industrial partners in Piedmont region (*Blue Engineering S.p.A., Italtecnica, De Marco S.p.A., Reply*).

DOCTORAL RESEARCHER

Politecnico di Torino

11/2021 - 10/2025

- Advanced CFD modeling of turbulence-combustion interactions in low-carbon fuel engines. **CONVERGE CFD** employed for 3D U-RANS simulations using **RNG k- ϵ** turbulence model. Validation against engine data provided by *FPT industrial*.
- Development of a new optimized strategy for **boundary condition setting** in 3D-CFD simulations of engines. Development of a new **spark deposition model for SI engines** in CONVERGE.

- **CINECA HPC Galileo100** cluster and **Polito HPC** efficiently used (considering Amdahl law and resources availability).
- **Time Frequency Analysis** of combustion to set-up a fast predictive real-time model in **Matlab** for engine diagnostics. In partnership with *FPT Industrial*.
- **Functional design** of a next-generation freight train with distributed power. **System Composer** has been used for architecture design and requirement management. **Simulink** for simulating and validating the Integrated Monitoring System. **GaBi** used for **Life Cycle Analysis** considerations. In partnership with *Alstom S.A.*.
- Simulating the operation of a next-generation freight train and its integration in intermodal logistic chains. **OpenTrack** and **Trenissimo** have been used for microsimulation of railway lines and terminals. **Anylogic** has been used for integrating microsimulation inside the processes of container handling at terminals. Use of **Galileo High Accuracy Service** for satellite-enhanced operation has been validated. **ARENA Discrete Events Simulation** has been used for the entire combined logistic chain simulation. In collaboration with *RFI* and *SITO interporto di Torino*.
- Coded a microsimulation tool for railway in **Python** for comparing different powertrain technologies for trains. The novelty consisted in automatically reconstructing the railway line through **OpenRailwayMapAPI** and **Google Maps API**
- **techno-economic analysis**, in partnership with *PSA Port of Genova*, of electric yard tractors.
- Analysis on energy consumption of battery electric buses for Public transport at different temperatures. **Data analysis** on fleet monitoring by *Gruppo Torinese Trasporti*.
- Thresholds conditioning scalability of electric vehicles with the application of **queueing theory** at public charging stations.
- Large-scale mobility pattern analysis and powertrain optimization. Processed **200k-vehicle European dataset** to identify urban/extra-urban driving cycles. Applied **k-means clustering** for grouping vehicles by usage patterns (15+ features), enabling targeted powertrain redesign. Data provided by *Stellantis*

TEACHING ASSISTANT

09/2022 - 03/2026

- Taught **120+ hours of frontal classes** on Fluid Machinery, Model Based System Engineering and Transportation Systems in **MSc courses** of Mechanical and Civil Engineering.
- Top scores (avg. 3.89/4) on students feedback questionnaires on lecturers performances for 3 years.
- **Supervised 20+ MSc students** for their final thesis in Mechanical, Civil, Energetics or Aeronautical Engineering.

PEER REVIEW ACTIVITIES

2022 - present

- **AIIT TIS Conference Rome 2022** (Transport Infrastructure & Systems)
- **AIIT TIS Conference Rome 2024** (Next-Gen Mobility Systems)
- **SAE Conference Torino 2024** (CO₂ Reduction Technologies)

CONTRACT PROFESSOR

Università della Calabria

07/2024

- Delivered course titled "Theory and Numerical Applications through Direct Data Collection from Vehicles" as part of the **postgraduate II level Master's program** in Electric Mobility and Circular Economy (MEES)

Professional Engineer

Italy

Ordine degli Ingegneri della Provincia di Torino

2022 - Present

- Licensed and qualified to practice as Industrial Engineer. Registered in the Order of Engineers of Turin, Reg. No. 14648.

RESEARCH SCHOLAR

Blue Engineering S.p.A. & Politecnico di Torino

07/2021 - 11/2021

- Conducted comprehensive analysis on advanced design and maintenance strategies for commuter rolling stock. Employed **Reliability Centered Maintenance (RCM)** principles to develop recommendations for implementing **predictive maintenance techniques** with **data-driven approaches** to improve operational efficiency.

PROCESS ENGINEERING INTERN

Alstom S.A., Savigliano (Italy)

09/2020 - 03/2021

- Implemented **Process FMECA** for Coradia Stream regional train validation, conducting **fault tree analysis** on 15+ critical subsystems. Developed **statistical process control** protocols improving testing efficiency. Collaborated with cross-functional teams to establish **risk priority number (RPN)** matrices for 200+ components, enhancing system reliability metrics.

ADDITIVE MANUFACTURING DESIGNER

Team PACE @PoliTo

09/2017 - 10/2018

- Part of an international team, comprising students from Politecnico di Torino, Tongji and Hunan Universities (China), TU Darmstadt and RWTH Aachen (Germany), and Kookmin University (South Korea). Secured *First Place in Engineering and Industrial Design* at the 19th PACE Forum competition held in Warren, Michigan, USA. More here: PACE.

Journal Publications

- 2025 De Marinis F., **Gurri S.***, Dabove P., *Development of a GNSS positioning system to support automatic manoeuvring of new-generation freight trains at low speeds*, *Ingegneria Ferroviaria* - PUBLICATION HERE: DOI
- 2025 **Gurri S.**, Cappelli F.*, Dalla Chiara B., *Matching demand and supply in motorised mobility: a data-driven differentiation of the driving patterns for urban contexts*, *Transportation Engineering* - 2025, PUBLICATION HERE: DOI
- 2024 Ferrari A., **Gurri S.**, Vento O.*, *Injected Fuel Mass and Flow Rate Control in Internal Combustion Engines: A Systematic Literature Review*, *Energies* - 2024, PUBLICATION HERE: DOI
- 2024 Olivari E., **Gurri S.**, Caballini C.*, Carotta T., Dalla Chiara B., *Ports go green: an energy-based cost analysis for evaluating the electrification of yard tractors*, *The Open Transportation Journal* - 2024, PUBLICATION HERE: DOI
- 2023 **Gurri S.***, Bocchieri M., Galasso D., Operti V., Dalla Chiara B., *Assessing the speed of an electric multiple-unit freight train on high-speed lines*, *Ingegneria Ferroviaria* - 2023, PUBLICATION HERE: DOI

Conference Contributions

- 2025 Olivari E., **Gurri S.***, De Marinis F., Caballini C., *A Hybrid Simulation Framework Combining DES and Microsimulation for Multicriteria Evaluation of Intermodal Freight Systems: A Case Study on Next-Generation Trains*, *TransClim 2025 (Transport Systems and Climate Change)*, June 18, 2025 (Livorno, Italy) - NO PROCEEDINGS
- 2024 **Gurri S.***, Zara G., Di Paola A., Dalla Chiara B., *Design of an integrated monitoring system for multiple-unit freight trains*, *6th International Conference on Railway Technology*, September 1-5, 2024 (Prague, Czech Republic) - PUBLICATION HERE: DOI
- 2024 **Gurri S.**, Cappelli F.*, Dalla Chiara B., *Matching demand and supply in motorised mobility via data analysis*, *SIDT Conference*, June 12-15, 2024 (Villasimius, Italy) - NO PROCEEDINGS
- 2024 **Gurri S.***, Olivari E., Caballini C., Dalla Chiara B., *Next-generation freight intermodal trains: an effective way to increase rail share and sustainability of inland transportation*, *SIDT Conference*, June 12-15, 2024 (Villasimius, Italy) - NO PROCEEDINGS
- 2023 **Gurri S.***, Zara G., Di Paola A., Dalla Chiara B., *Preliminary design of an integrated monitoring system for multiple-unit freight trains*, *UIC high speed rail - 07-10 March 2023 Marrakesh*, March 07-10, 2023 (Marrakesh, Morocco) - NO PROCEEDINGS
- 2022 **Gurri S.***, Santacaterina E., Guarrera M., Dalla Chiara B., *Driving modal shift on low-traffic railway lines through technological innovation*, *Transportation Research Procedia - AIIT International Conference TIS*, September 15-16, 2022 (Rome, Italy) - PUBLICATION HERE: DOI
- 2022 **Gurri S.***; Bocchieri, M.; Galasso, Daniela; Operti, V.; Dalla Chiara, B., *Simulating the operation of new-generation freight-EMUs for high-speed lines: perspectives for more reliable, sustainable, and fast logistics*, *5th International Conference on Railway Technology*, August 22-25, 2022 (Montpellier, France) - PUBLICATION HERE: DOI

Under Review

- 2026 **Gurri S.**, Vento O.*, Ferrari A., *Investigation of cyclic variability in spark ignition internal combustion engines with a physical spark discharge model for RANS 3D-CFD simulations*
- 2025 **Gurri S.**, Vento O.*, Marello O., Finesso R., Gallo A., Vargiu L., Ferrari A., *A Methodology for boundary condition optimization in 3D-CFD of internal combustion engines*
- 2025 Olivari E., **Gurri S.***, De Marinis F., Caballini C., *A simulation-based decision-making framework for sustainable intermodal transport including next-generation freight-EMUs*
- 2025 Sardella G., **Gurri S.***, Andrenacci N., Dalla Chiara B., Vellucci F., Pede G., Zanini M., *BEB for Public transport: a methodological approach and data resulting on energy consumption at different temperatures*

Dissemination

2023 Dalla Chiara, B.; Gurri, S., *Efficienza energetica nel trasporto merci: quali strumenti.*, *Logistica Management*, pp 54-61 - MARCH 2023

HARD SKILLS

CFD/Thermodynamics	Converge CFD, GT Power, Cantera
CAD	Solidworks, Inventor, Autocad 2D, NX11
Discrete Event Simulation	Arena, Anylogic
Railway Microsimulation	OpenTrack, Trenissimo
Coding and data analysis	Python, Matlab, Excel
Others	HPC systems (MPI, OpenMP)

COURSES AND CERTIFICATIONS

- 2025 **AI/ML for Fluid Dynamics**
22 hours, *Flow Thermo Lab*
- 2024 **Dive into the World of HPC: Parallel and Accelerated Computing**
80 hours, *CINECA*
- 2023 **Towards Data Analysis and Machine Learning – Data Structures in Python (Advanced Level)**
40 hours, *Politecnico di Torino*
- 2021 **Railway Engineer Course – Organization, Technics, and Safety in the Railway Field**
60 hours, *CIFI*
- 2021 **Graduate Record Examination (GRE) – General Test**
ETS
- 2019 **TFI – Test de Français International**
ETS EMEA
- 2018 **CSWA – Certified SOLIDWORKS Associate in Mechanical Design**
SolidWorks Authorized Training Centre

HOBBIES AND ICE BREAKERS

Outdoorsy adventures	Hiking and running, with forays into mountain biking, all around the World. Balancing physical challenges with mindful practices like yoga, embracing both the thrill of exploration and moments of tranquility in nature
Sport endeavors	Watching and playing sports give salt to every day. Be it padel, tennis, volleyball or handball, I thrive on the energy of competition and the joy of teamwork. Pushing my limits on the field while maintaining perspective: finding that sweet spot where determination meets enjoyment, where every match brings both challenge and laughter.
Storytelling	Amateur photographer of urban and wild landscapes, capturing the interplay of light and shadow and of the different colors of nature. Combining photos with creative writing to craft compelling narratives to share with friends.
Continuous learning	Voracious reader of essays and psychological literature. Actively engaged in online courses on emerging technologies, bridging theoretical knowledge with practical applications. Always trying to improve my communication skills in as many languages as possible.